Prefill Once, Ask Many: Training-Free Reversible Below-Page KV Selection for Multi-Query Long-Document Visual Question Answering

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Abstract

Long-document visual question answering is naturally multi-query: a document is read once, but successive questions need different evidence. Existing KV-cache eviction methods commit to one compressed context, so evidence discarded for the first question is unavailable to later ones. We present PAQ-KV ("Prefill once, Ask many Queries"), a training-free method that prefills a document once, retains its complete KV cache, and constructs a new query-specific attention view for every question: the frozen reader's own question-to-context attention scores 64-token page-aligned blocks, and decoding applies a four-dimensional additive mask that exposes only the selected blocks at their original M-RoPE positions. Because the cache is neither gathered nor evicted, every selection is reversible, with no additional model, retrieval index, or document re-read. At a nominal 5% decode-attention budget ($\approx$6% realized), PAQ-KV scores 0.404 on MMLongBench-Doc versus 0.328 for the trained ColQwen2 retriever, and a disjoint sealed test set evaluated exactly once under a frozen protocol confirms the improvement ($+0.076$, document-clustered 95% CI $[+0.029, +0.122]$). Per-query re-selection also outperforms committing the identical scorer once, by wide margins. Further datasets and readers bound the scope: the advantage is largest where evidence is dispersed and the host reader's attention localizes it well, and reverses on a retrieval-native cell and on InternVL3.5-8B.

1 Introduction

Long-context vision-language models (VLMs) can read an entire multi-page document, but they pay for every page on every question. The standard efficiency response, inherited from single-query text benchmarks, is to *evict*: after reading, the model keeps only a compact subset of its KV cache, the stored per-token state that decoding attends to, and discards the rest (Li et al. 2024; Zhang et al. 2023; Chen et al. 2024; Wan et al. 2024; Kim et al. 2025). Eviction is a single, irreversible commitment: whatever subset survives must serve every future question about that document.

In practice, however, document assistants are not asked one question. Both benchmarks in this study, MMLongBench-Doc (MMLB; Ma et al. 2024) and LongDocURL (LDU; Deng et al. 2025), pose long-document visual question answering (VQA) natively as groups of questions about a shared document. Inside our pinned evaluation subsets the average document carries more than five gold-bearing questions (§4). Different questions need different evidence, so a subset committed for question 1 is stale by question 5.

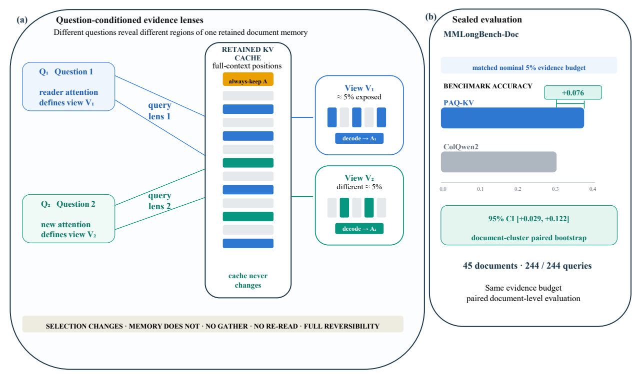

Figure 1: Method and sealed result at a glance. On fresh, disjoint MMLB, PAQ-KV@5% scores 0.396 versus 0.320 for corrected-loader ColQwen2 ($+0.076$, document-clustered 95% CI $[+0.029, +0.122]$; 45 documents, 244/244 queries).

This paper asks two questions. At a matched evidence budget on long multimodal documents, does *re-selecting* context for every query beat selecting once and committing? And can a training-free selector built from the reader's own internals compete with a dedicated trained retriever? Our answer is PAQ-KV (Prefill once, Ask many Queries).

PAQ-KV works as follows. A frozen reader (Qwen3-VL-8B; Qwen Team 2025b) prefills the document once and keeps the resulting KV cache. Each question is then scored against that cache by the reader's own attention from the question tokens to the document tokens, and those scores rank the cache's 64-token blocks. Blocks are the granularity below the page: a typical page spans several of them, so a block is roughly a table row, a figure caption, or a paragraph rather than a whole page of mixed content. The answer is decoded from the full cache under a four-dimensional attention mask that exposes only the top-ranked blocks. Nothing is gathered and nothing is discarded, so every position index stays the model's own. This matters because the reader family indexes tokens with M-RoPE (Qwen Team 2025a), a multimodal rotary position embedding that gives an image token its temporal, height, and width coordinates on the page rather than a flat sequence offset; physically compacting the cache would have to reassign these composite indices, while leaving it in place avoids the question entirely. Each selection is therefore *reversible*: the next question re-selects from the same intact cache. Figure 1 summarizes the architecture and its sealed MMLB result.

Contributions.

- **A reversible, training-free selection architecture.** PAQ-KV retains the full-document KV cache and gives every query its own below-page selection through a per-query 4D additive decode mask, preserving native M-RoPE positions with no gathering, eviction, or re-read (§3).
- **A pre-registered isolation of reversibility.** Holding the scorer and budget fixed, per-query re-selection outperforms the same scorer committed once by $+0.137$/$+0.211$ (MMLB/LDU) and a gold-informed static subset by $+0.213$/$+0.230$, with document-clustered intervals excluding zero in every cell (§5.1).
- **An out-of-sample-confirmed improvement over a trained retriever, with its scope measured.** At a matched 5% budget PAQ-KV outperforms ColQwen2 by $+0.077$ on MMLB, confirmed once on a sealed disjoint set ($+0.076$); MMDocRAG reproduces the improvement, a retrieval-native MMDocIR cell reverses it, LDU is a statistical tie, and a full-method port to three further readers bounds generality (§5).

2 Related Work

Page selection and retrieval. CAPS (Zhu et al. 2026) publishes attention-based *page* selection on our benchmarks, and ColPali/ColQwen2 (Faysse et al. 2025) are the strongest trained retrieval baselines we compare against. Selecting pages with reader attention is CAPS's method; PAQ-KV differs on the axes CAPS names as open. It selects below the page, it re-selects per query over a single retained prefill rather than committing once, and it decodes directly from the retained cache with no re-read.

KV-cache eviction. SnapKV (Li et al. 2024), H2O (Zhang et al. 2023), FastV (Chen et al. 2024), LOOK-M (Wan et al. 2024), and KVzip (Kim et al. 2025) all commit a compressed cache once. Quest (Tang et al. 2024) shows that query-aware selection can outperform query-agnostic eviction in single-query text, which motivates but does not prove our multi-query claims. We did not find prior work that combines reader-internal attention as the runtime below-page selector, end-to-end question answering at a matched budget against page-level retrieval, and long multi-query documents; we offer this as a provisional reading of the literature rather than proof of novelty.

3 Method: PAQ-KV

3.1 Setting

One long visual document, consumed as page images with no OCR, is read by a frozen VLM and will be asked k questions in sequence. The document is prefilled *once*, at the same rendering the full-context arm uses and a nominal 64K-token context target (§4 gives realized lengths), and the KV cache is retained: never evicted, never physically compressed, never gathered. The cache plays the role that a multi-vector index plays for a retriever, except that it is the reader's own working memory, so retrieval and reading share one representation and one model.

3.2 Per Query: Preview, Re-Select, Masked Decode

1. **Preview (score).** The tokenized question is forwarded over the retained cache, and the attention from the question rows to the context positions is aggregated into scores over 64-token, page-aligned KV blocks. Blocks at this size are fine enough to isolate a table row, a figure caption, or a paragraph, yet coarse enough for their scores to be stable. After the preview the cache is cropped back to the document boundary, so one query never contaminates the next (§4.4).
2. **Re-select.** The top-scoring blocks are kept, up to a nominal 5% decode-attention budget. The term is deliberate: the kept blocks' keys and values were computed with full-document context during the prefill, as is standard in the Quest/SnapKV method class, so the budget bounds what decoding may attend to rather than what prefill may read. The realized attended fraction runs slightly above nominal, $\approx$5.9 to 6.0% of context on the executed screen, and is reported alongside every result. Against ColQwen2 the matched quantity is the same 5% evidence budget, counted in pages for the retriever and in context tokens for PAQ-KV, under one shared reader and answer evaluator; the two methods differ only in how they select.
3. **Masked decode.** The answer is generated while attending only to the selected blocks plus a small always-keep set: the scaffold of BOS, system, and template tokens, the question itself, and previously generated tokens. The restriction is enforced by a four-dimensional additive attention mask over batch, head, query row, and key, applied over the retained cache. There is no gathering: the K/V tensors stay at their original positions, so every M-RoPE rotary index is exactly the model's own, and the failure mode that makes physical cache-gathering error-prone for VLMs does not arise. There is no re-read, no re-render, no second model.

Formal definition. Let Q_j be the token positions of question j, let V be the visual-content positions, and let $B_1, \ldots, B_m$ be the page-aligned blocks partitioning V. For attention probability $A_{\ell h}(r,t)$ from question row r to context position t at layer ℓ and head h, the default scorer averages over all L layers, H heads, and question rows, then averages within each block:

$$\alpha_j(t) = \frac{1}{LH|Q_j|} \sum_{\ell=1}^{L} \sum_{h=1}^{H} \sum_{r \in Q_j} A_{\ell h}(r,t),$$
$$s_j(B_i) = \frac{1}{|B_i|} \sum_{t \in B_i} \alpha_j(t). \tag{1}$$

Let π_j order blocks by decreasing s_j. At budget b, selection stops at the first block whose inclusion reaches the visual-token budget:

$$r_j = \min\left\{ r : \sum_{u=1}^{r} |B_{\pi_j(u)}| \geq b|V| \right\}, \qquad S_j = \bigcup_{u=1}^{r_j} B_{\pi_j(u)}. \tag{2}$$

If $\mathcal{A}$ denotes the always-keep scaffold and G_j the generated-token positions, define $T_j(x) = \mathcal{A} \cup S_j \cup \{t \in Q_j \cup G_j : t \leq x\}$. The additive mask for decode row x is

$$M_j(x,t) = \begin{cases} 0, & t \in T_j(x), \\ -\infty, & \text{otherwise.} \end{cases} \tag{3}$$

This row-wise mask is broadcast over batch and heads. Decoding likewise ends with the cache cropped back to the document boundary, so S_{j+1} is constructed from the same retained cache.

Algorithm S1 (Suppl. §A.1)[1] gives the loop and Figure 2 shows the mask machinery. During the preview, attention is captured by per-layer forward hooks: each layer's attention tensor is reduced immediately and then discarded, so peak memory holds one layer's attention at a time. Retaining all layers' tensors, the naive implementation, runs out of memory at long contexts.

3.3 Exactness and Reversibility

A decode-identity audit, pre-registered as a kill criterion, checked the mask path against the native decode on six document-query pairs spanning 32K to 122K tokens: in all six cases an all-true mask produced zero mask-specific argmax changes, so we claim identity in the audited cases, not in general (Suppl. §C.2 gives the audit detail). Because nothing is discarded, every selection is furthermore *reversible*: the next query re-scores and re-masks the same complete cache, whereas whatever commit-once eviction dropped is unrecoverable. Reversibility is the axis our evaluation isolates (§4.2, §5.1).

Relation to existing designs. PAQ-KV is not CAPS (a separate scoring VLM, per-page full re-forwards, page granularity, single-query); it is not coarse-to-fine page routing with a full-resolution re-read (PAQ v1, our own earlier page-level system, Suppl. §C.5); and it is not eviction (the mask is per-query and reversible). The mask route is the *accuracy* path; physically gathering the cache for savings is a separate efficiency claim, and one we do not make here.

4 Experimental Setup

4.1 Benchmarks, Pins, Reader

We evaluate on LongDocURL (LDU; Deng et al. 2025) and MMLongBench-Doc (MMLB; Ma et al. 2024), both as page images, and add two development-grade cells built from MMDocRAG (Dong et al. 2025b) and MMDocIR (Dong et al. 2025a) (§5.3). Each cell uses a fixed, seed-frozen evaluation subset (a *pin*): the confirmatory pin holds 80 documents per cell, with 463/440 complete queries and 5.79/5.50 gold-bearing queries per document (LDU/MMLB), each document prefilled once at a nominal ≈64K-token context so every query's evidence is present in one prefill; Suppl. §A.1 gives the construction rules. A 30-document screen pin is its strict prefix, and results are additionally checked on the 50 new documents only (§5.1). The reader is Qwen3-VL-8B (Qwen Team 2025b), frozen throughout: scaled dot-product attention serves prefill and reading, and eager attention is enabled only during the per-query preview so question-to-content attention can be captured. Page-level arms operate at a matched page budget $b \in \{2.5\%, 5\%, 10\%\}$, with the pre-registered gate budget at $b = 5\%$; accuracy is the benchmark's document-VQA scorer. The primary results are a Qwen3-VL-8B case study; §5.3 ports the full method to three further readers as a development-pin generality control.

Evidence status. Primary accuracy results use the 80-document-per-cell development pins unless labelled otherwise; mechanism, budget, and integrity analyses use smaller screening subsets whose sizes are stated where used. *Sealed* here means a disjoint held-out test set evaluated exactly once under a frozen protocol: only the MMLB PAQ-KV-versus-ColQwen2 comparison was sealed-evaluated, on a fresh 45-document cell (§5.2); the sealed LDU documents remain unevaluated. The cross-reader results, the MMDocRAG and MMDocIR-QA cells, the same-cohort baseline panels, and the MMDocIR selection-quality analysis are development-grade additions with single runs and no pre-frozen bars (§5.3; Suppl. §B.5 and §B.8).

4.2 Statistical Protocol

All inference uses a document-cluster paired bootstrap: queries of one document share the document and (for stale arms) the single committed subset, so bootstrapping over queries would be anti-conservative pseudoreplication. Intervals resample documents with replacement within each cell (10,000 draws, seed 0). Paired contrasts are reported as the mean difference Δ with its two-sided 95% confidence interval (CI), written $\Delta\ [l, u]$; bare bracketed pairs after an effect denote this interval throughout. Every pre-registered gate in this study is a criterion on the interval's lower bound: a gate passes when the lower bound exceeds zero, or the stated margin; where that bound alone carries the argument we abbreviate it LB_{95}. Per-cell results are primary; pooled contrasts use equal cell weighting and are labelled. Every gate was frozen before its run.

The isolation gate. That query-aware selection outperforms query-agnostic eviction is both a known text result (Tang et al. 2024) and a potential strawman. We therefore fix the tested axis to *reversibility*: every baseline runs its canonical scorer but is deployed stale, committed once per document (query-aware scorers commit a min-max-normalized aggregate over the document's first $M = \min(3, k)$ queries),

[1]References of the form Suppl. §X, Table SX, Figure SX, and Algorithm SX point to the separate supplementary-material document.

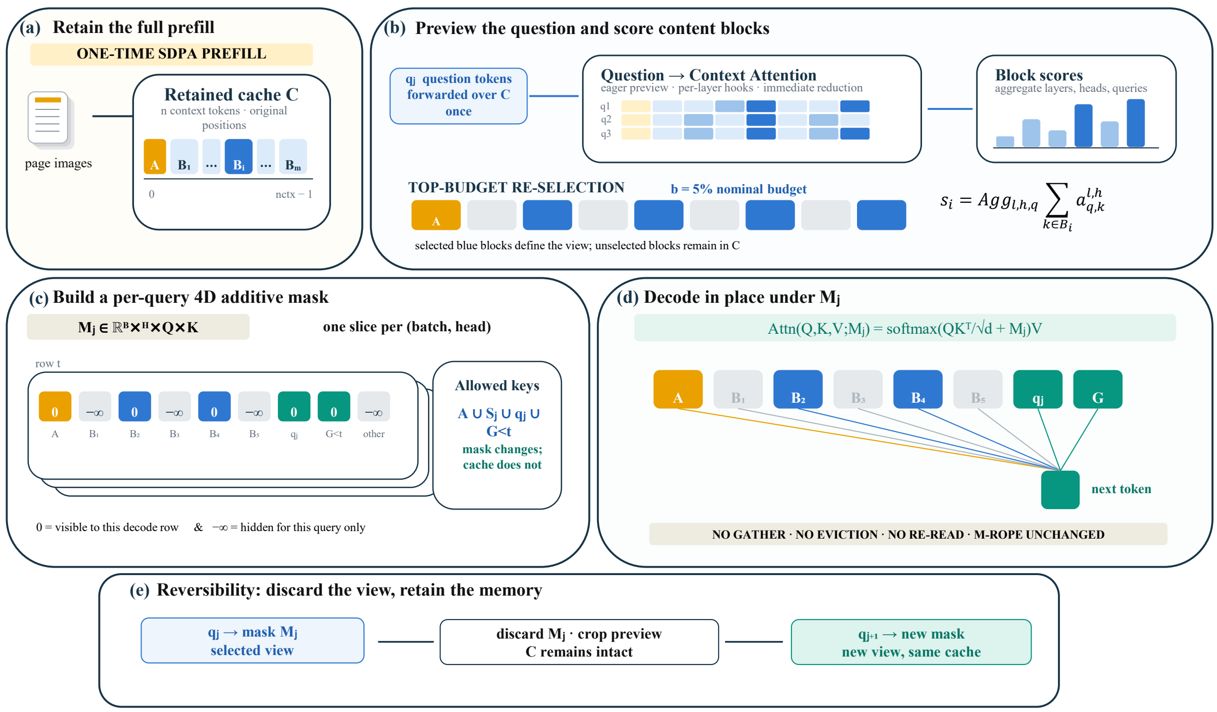

Figure 2: Detailed PAQ-KV restricted-attention mechanism. **A:** The frozen reader prefills the document once and retains the full KV cache C, including the always-keep scaffold $\mathcal{A}$ and page-aligned 64-token blocks $B_1, \ldots, B_m$, at their native M-RoPE positions. **B:** For each question q_j, an eager-attention preview captures question-to-context attention using per-layer hooks. Attention mass is aggregated into block scores, and the highest-scoring blocks S_j are retained under the nominal 5% decode-attention budget. The preview is then cropped so that questions cannot contaminate subsequent queries. **C:** PAQ-KV constructs a query-specific additive mask $M_j \in \mathbb{R}^{B \times H \times Q \times K}$, assigning 0 to $\mathcal{A} \cup S_j \cup q_j \cup G_{<t}$ and $-\infty$ to all other keys. **D:** Decoding proceeds directly over the intact cache under M_j, without gathering or evicting K/V, re-reading the document, or changing positional indices. After answering, M_j is discarded, and the same complete cache is re-masked for q_{j+1}.

while the fresh arm re-selects per query. Reversibility is credited only if the fresh arm outperforms two comparators. The first is its *own* scorer committed once: scorer and budget are identical, so the delta isolates the mechanism. The second is a strong *gold-informed* static subset, the top pages by gold-page frequency across the document's queries: not deployable, and not provably optimal, but immune to the weak-static artifact that invalidated an earlier one-comparator gate. The gate is executed with PAQ-KV itself as the fresh arm at its native below-page granularity (the committed aggregate and the gold-frequency scores are applied over the same 64-token blocks), on the full 80-document pins of both cells, post-fix code only; the bar, arms, budget, seed, imputation policy, and code hashes were frozen and committed before the run (Suppl. §A.2).

4.3 Baselines

The executed panel (Suppl. §B.7) compares, per query at matched budget on the same pinned data and reader: ColQwen2 (Faysse et al. 2025) fresh and stale, CLIP (Radford et al. 2021) fresh, SnapKV fresh and stale, the isolation arms above, random and truncation controls, and a query-blind eviction family (H2O, a context-only FastV proxy, LOOK-M, and a KVzip recv_max proxy). The eviction arms are disclosed proxies that sit at or below random page recall on these cells, so the trained retriever is the binding comparison; we rest no claim on outperforming proxy eviction. Full-context and gold-page-oracle rows are references, not competitors. The training-free arm family is executed at matched page-set semantics on the development reader for every cell of Table 1 and on all three second readers (Suppl. §B.5).

4.4 Methods Integrity

Two independent code reviews caught defects that we repaired and quantified rather than assumed away; Suppl. §A.3 gives both accounts in full. First, the per-query scoring loop reused the document prefill without restoring it, so early queries contaminated later previews. The fix, cropping the cache after every preview, is applied everywhere; every gate and headline number postdates it, the confirmatory isolation gate was re-executed in full on post-fix code with runtime verification, and the key isolation contrast is essentially unchanged (+0.264 clean versus +0.284 polluted on a 24-document re-run). Second, the cached ColQwen2 baseline had loaded with its trained final retrieval-projection

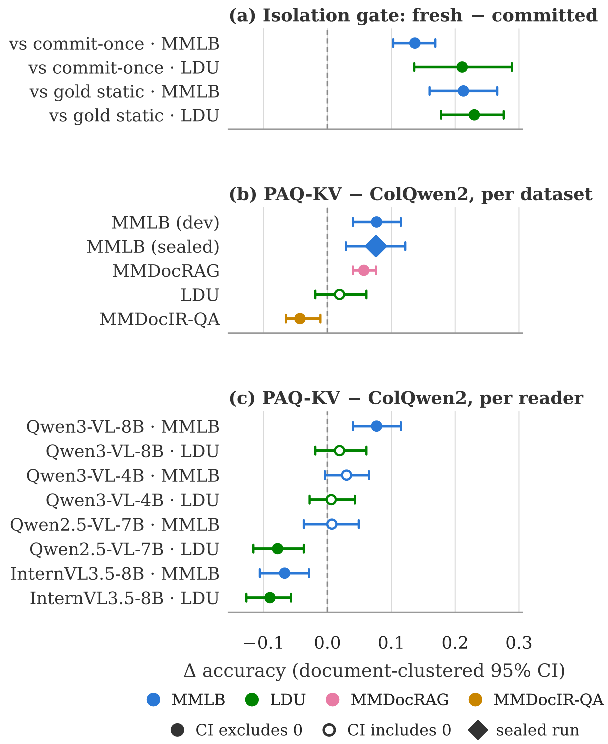

Figure 3: Key paired contrasts (mean Δ accuracy, document-clustered bootstrap 95% CIs); the key below the panels gives the benchmark color coding and marker semantics. **(a)** The confirmatory isolation gate: PAQ-KV minus its own scorer committed once and minus a gold-informed static subset (§5.1). **(b)** PAQ-KV minus ColQwen2-fresh per dataset on Qwen3-VL-8B (§5.2, §5.3). **(c)** The same contrast per reader on the development pins (§5.3). Exact values in Suppl. §B.1 and §B.4.

LoRA silently zeroed. We patched the loader, added a post-load assertion, and re-scored the comparator: the correction is immaterial, with the MMLB improvement at $+0.077$ against both loaders and the LDU contrast at $+0.019$ against both. The corrected loader is used everywhere; the one remaining pre-fix artifact is the historical page-level panel of Suppl. §B.7 ($\approx$0.04 fresh-arm inflation), whose uncorrected-loader retrieval rows are omitted in favor of Table 1's same-cohort corrected values and on whose absolute levels no claim rests.

Separately, the study adopted a sealed-test discipline: the 80-document pin is development-only, and a fresh, disjoint 45-document / 244-query MMLB set was reserved and evaluated *exactly once* under a frozen, hash-guarded harness (Suppl. §A.2). That one sealed evaluation confirms the improvement out-of-sample (§5.2); the sealed LDU documents were not evaluated.

5 Results

5.1 Does Per-Query Re-Selection Matter?

The isolation gate runs at $b = 5\%$ on all 907 pinned queries of the confirmatory pin (80 documents per cell; zero skips), pitting PAQ-KV itself, per-query below-page re-selection over the retained cache, against two comparators built from the same 64-token block scores. PAQ-KV scores 0.531/0.405 (LDU/MMLB), the identical scorer committed once scores 0.320/0.268, and the gold-informed static scores 0.301/0.192. Both gate contrasts pass in both cells, by $+0.211/+0.137$ over commit-once and $+0.230/+0.213$ over the static (Figure 3a; full table in Suppl. §B.3). The committed subset is not a strawman: it shares a third of the fresh arm's blocks (mean Jaccard 0.368) and still loses, and all three arms realize a matched $\approx$6.0% of context. The contrast survives three robustness checks (Suppl. §C.2): re-running the statistics on only the 50 documents outside the screen prefix ($+0.180/+0.206$ pooled), a clean-cache page-level re-run ($+0.264$, LB_{95} $+0.187$), and a scorer swap in which ColQwen2 itself collapses when committed once (0.423 fresh versus 0.178 stale), so reversibility is a property of the multi-query regime rather than of our scorer. A pre-registered "commitment-pressure" mechanism gradient failed its formal interaction test and is retained as exploratory only.

5.2 Does PAQ-KV Compete with Trained Retrieval?

Main comparison. On the 80-document development pin (444 queries decoded; contrasts on the 440 with cached references), PAQ-KV@5% scores 0.404 versus 0.328 for ColQwen2-fresh. The improvement of $+0.077$ $[+0.040, +0.115]$ meets the pre-registered per-cell bar with margin (Figure 3b; Table S2 in Suppl. §B.1). ColQwen2 here runs with its trained retrieval-projection LoRA correctly applied; §4.4 documents the loader defect we found and corrected and shows the correction is immaterial to this comparison. Two development-pin re-slices sharpen where the improvement comes from. At the same nominal budget the retriever delivers *more* gold evidence to the reader, not less: ColQwen2's kept pages cover 0.539 of gold-page tokens versus 0.344 for PAQ-KV's selected blocks, yet PAQ-KV answers more accurately, so gold-evidence recall and answer accuracy dissociate. And the paired improvement concentrates on cross-page evidence ($+0.109$ $[+0.052, +0.170]$, versus $+0.054$ $[+0.001, +0.108]$ on single-page evidence), which directly supports the dispersed-evidence reading of this cell. Suppl. §C.3 gives both recall notions and the full evidence-type stratification.

Out-of-sample confirmation. The improvement holds on the sealed set (Figure 4). On 45 MMLB documents / 244 queries with zero overlap with the development pin, under a harness whose bar, comparator, budget, imputation policy, and code hashes were frozen before the run (Suppl. §A.2), PAQ-KV@5% scores 0.396 versus ColQwen2-fresh 0.320: $+0.076$ $[+0.029, +0.122]$, complete (244/244), unchanged under worst-case missingness imputation, and closely matching the development estimate

	LDU 80 docs / 463 q	MMLB 80 docs / 444 q	MMDocRAG 62 docs / 585 q	MMDocIR-QA 80 docs / 538 q
PAQ-KV@5% (**ours**)	**0.5315**	**0.4042***	**0.4089***	0.3654
ColQwen2-fresh (matched)	0.5121	0.3282	0.3517	**0.4080***
CAPS-style expert head (page)	0.5182	0.3561	0.3667	0.3618
BM25 (page retriever)‡	0.4030	0.2109	0.2793	0.3522
SnapKV-fresh (page)	0.4892	0.3243	0.1557	0.1987
SnapKV-stale (commit-once)	0.1873	0.0902	0.0750	0.1228
CLIP-fresh	0.1549	0.1624	0.1119	0.1400
ColQwen2-stale (commit-once)	0.2160	0.1365	0.1749	0.1583
random	0.0675	0.0218	0.0368	0.0807
truncation	0.0329	0.0410	0.0327	0.0876
full-context (ref.)	0.5456	0.3965	0.4018	0.3785
gold-page oracle (ref.; not strict)	0.6092	0.4314	0.4028	0.4608

Table 1: Main results: accuracy at the nominal 5% budget per benchmark (reader Qwen3-VL-8B). Bold marks the best non-reference arm per column; * marks a paired PAQ-KV-versus-ColQwen2 difference whose document-clustered 95% CI excludes zero (Suppl. §B.1), placed on the better arm. LDU (LongDocURL) and MMLB (MMLongBench-Doc) are the 80-document development pins; MMDocRAG and MMDocIR-QA are development-grade cells with single runs and no pre-registered bars. The sealed MMLB confirmation is deliberately not a column: it is shown in Figure 4 and Figure 3b. All baseline rows share PAQ-KV's post-fix frozen stack at matched page-set semantics. ‡BM25 text channels differ by corpus. Suppl. §B.2 gives realized budgets, text channels, and cohort notes.

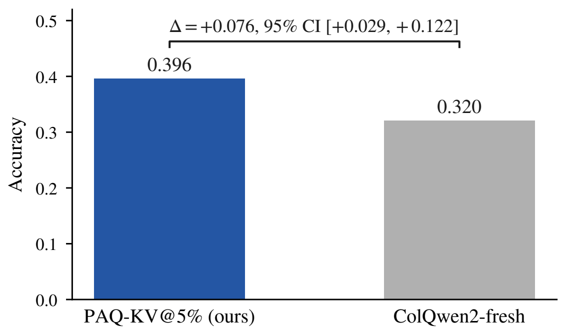

Figure 4: The sealed out-of-sample confirmation on MMLB: accuracy of PAQ-KV@5% and ColQwen2-fresh on the fresh, disjoint 45-document, 244-query set, evaluated exactly once under the frozen harness (§4.4). The bracket reports the paired difference with its document-clustered bootstrap 95% CI; only this comparison was run on the sealed set.

$(+0.077)$. Only this comparison was sealed-evaluated; the reference arms are development-pin only. This single evaluation is the strongest evidence in the paper and directly addresses the development pin's reuse across the study. On the development pin, PAQ-KV at the same budget is indistinguishable from full context $(+0.009\ [-0.022, +0.040])$ and not statistically distinguishable from the gold-page oracle $(-0.026\ [-0.059, +0.008])$; the latter interval permits material inferiority, so this is parity within noise rather than demonstrated equality. A training-distribution audit found no confirmed document-level overlap between the comparator's training mixture and our corpora, and confirms it is trained for exactly this task family on the same document genres (Suppl. §A.2).

Mechanism checks. On the 30-document screen the result survives every mechanistic adversarial check: selection is query-dependent $(+0.329$ over random blocks), query-specific $(+0.698$ over wrong-question selection), reversible $(+0.379$ over a first-query commit), and below-page granularity helps over page level $(+0.054\ [+0.011, +0.102])$. Suppl. §C.1 details these screen analyses and their caveats, including a screen-level budget-monotonicity failure that the full-pin sweep of Suppl. §B.6 attributes to small-sample noise.

Additional matched-budget baselines. Table 1 carries the full training-free family, re-run on PAQ-KV's own post-fix cohort at matched page-set semantics. SnapKV-fresh, the strongest training-free arm, scores pages from the same full-resolution prefill PAQ-KV uses and still trails it, by $+0.080$ $[+0.043, +0.119]$ on MMLB and $+0.042$ $[-0.001, +0.092]$ on LDU. A BM25 page retriever trails PAQ-KV by $+0.193$ and $+0.129$ $[+0.083, +0.176]$ respectively. The closest chaser is a training-free *page-level* CAPS-style expert-head control with per-cell cross-validated head selection: PAQ-KV exceeds it by $+0.048$ (LB_{95} $+0.009$) on MMLB and is at parity on LDU, so the improvement over the trained retriever remains the stronger claim. The full-pin budget sweep is in Suppl. §B.6; the query-blind eviction family remains in Suppl. §B.7, at or below random page selection. Two pre-registered extensions of the scorer failed their pre-frozen gates and are reported in Suppl. §C.4: a learned per-head attention mixture did not transfer, and a single sharp expert head did not improve on the diffuse block scorer.

5.3 Where Does PAQ-KV Help?

LDU: a statistical tie. On the 80-document development pin (463 queries), PAQ-KV@5% posts a positive point estimate over ColQwen2-fresh of $+0.019$ $[-0.019, +0.061]$, which does not meet the pre-registered bar (Figure 3b). On this cell PAQ-KV also sits below full context, with an interval that includes zero, and clearly below the gold-page oracle. The mechanism checks, query dependence and reversibility, pass here too; what is missing is the margin. We treat this as a bounding observation on where below-page selection helps rather than a second main result, and defer the conversion-bound decomposition to Suppl. §D.

MMDocRAG: the improvement reproduces. To test the dispersed-evidence reading beyond the two primary benchmarks, we added MMDocRAG (Dong et al. 2025b), a retrieval-augmented document-QA dataset with cross-modal, multi-page evidence chains (52% of queries carry cross-page evidence). MMDocRAG is corpus-adjacent to our pins, so a document-name audit ran first: every document overlapping our development or sealed pins was excluded before any experiment, leaving a fresh 62-document, 585-query development-grade cell (Suppl. §A.2 details the audit and this pin's protocol deviations). The improvement over the trained retriever reproduces: PAQ-KV@5% scores 0.409 versus 0.352, a margin of $+0.057$ $[+0.040, +0.076]$ in the same class as the MMLB development and sealed results. PAQ-KV is again statistically indistinguishable from its own full-context read and from the gold-page oracle, and the oracle here barely exceeds full context, which is what genuinely dispersed evidence predicts. The training-free baselines fall far behind, committing either scorer once collapses it here too, and PAQ-KV clears even the benchmark-supervised expert-head control ($+0.042$ $[+0.030, +0.055]$; Table 1). A query-cap check (Suppl. §A.2) confirms the capped cell did not manufacture the result.

MMDocIR-QA: a retrieval-native reversal. MMDocIR's questions carry reference answers as well as gold evidence, so the fresh-documents pin of Suppl. §B.8 (80 documents, 538 queries, same overlap audit) also yields a fourth, development-grade QA cell, the most retrieval-native of the four: its questions were written per annotated layout for retrieval evaluation. Here the trained retriever wins. ColQwen2-fresh scores 0.408 versus 0.365 for PAQ-KV, a significant deficit of -0.043 $[-0.065, -0.011]$, while PAQ-KV remains indistinguishable from its own full-context read and far above the training-free arms (Table 1). The cell rewards retrieval as such: plain BM25 over the benchmark's own text ties PAQ-KV, the supervised expert-head control ties it as well, the trained retriever wins outright, and, unlike the dispersed cells, the gold-page oracle sits well above full context, so the cell's evidence regime, not selector quality, sets the sign. Across the four cells the pattern is sharper than three positives would have been: PAQ-KV wins where evidence disperses (MMLB and MMDocRAG), ties where conversion binds (LDU), and loses where questions are retrieval-native and evidence is concentrated (MMDocIR-QA); Figure 3b carries all five contrasts. This is the scope condition of the conclusion, measured rather than asserted.

Cross-reader generality. All results above use one reader. To test whether the effect is specific to Qwen3-VL-8B, we ported the complete method (reader-own attention scoring, block selection, masked decode) to three further VLMs: Qwen3-VL-4B (the same architecture at half the parameters), Qwen2.5-VL-7B (Qwen Team 2025a) (the prior generation of the same family), and InternVL3.5-8B (Wang et al. 2025) (a different VLM whose language backbone is Qwen3). Every port passed the same all-true-mask decode-identity gate before any accuracy number was read. The outcome is a gradient ordered by architectural distance from the development reader (Figure 3c; full table and protocol in Suppl. §B.4). Qwen3-VL-4B largely preserves the effect ($+0.030$ $[-0.004, +0.065]$ on MMLB) and, like the 8B, scores above its own full context. Qwen2.5-VL-7B is a statistical tie ($+0.007$ $[-0.037, +0.049]$). InternVL3.5-8B falls significantly below both its comparator (-0.067 $[-0.106, -0.029]$) and its own full context. On LDU every reader sits at or below its MMLB contrast with the ordering preserved, consistent with that cell being conversion-bound. A pre-registered development-only tuning study on InternVL found budget to be the only effective lever: the deficit narrows monotonically to -0.001 at a 30% budget, but the lower bound never clears zero, so the negative is not an artifact of the 5% budget or of the attention read-out (Suppl. §B.4).

The gradient tracks a measurable property: how well the host reader's own attention retrieves evidence below page level. On the queries a reader answers correctly from its own full context, its 5% selection recovers 0.88 (Qwen3-VL-8B), 0.87 (Qwen3-VL-4B), 0.70 (Qwen2.5-VL-7B), and 0.53 (InternVL3.5-8B) of that accuracy. Since the effect survives halving the reader within the same architecture but reverses across families, we read the boundary as architectural rather than a matter of scale, and state the scope condition plainly: PAQ-KV is a white-box method whose benefit is contingent on the host reader's attention being an effective below-page retriever. PAQ-KV nonetheless stays ahead of every training-free baseline on every reader, including those where it loses to the trained retriever (Suppl. §B.5). Three caveats attach: these are development-pin generality controls, not sealed claims; the two non-Qwen3-VL readers run at reduced realized context, so only within-reader contrasts are interpreted; and InternVL3.5-8B's language backbone is Qwen3, so it is a cross-VLM test but not a fully independent language architecture. Three further candidates could not run the method under the pinned software stack and are reported as incompatible rather than forced (Suppl. §B.4).

5.4 Systems Measurements

We measured the deployment profile of the true PAQ-KV path (below-page masked decode over the *retained* full-document KV cache) against ColQwen2-fresh and full context on an 8-document, 31-query systems subset of the pin (single GPU, frozen configuration; Table 2). PAQ-KV is roughly $2.25\times$ slower per online query than the retriever pipeline and retains a multi-gigabyte per-document cache where the retriever stores a megabyte-scale page index, but it is roughly $11\times$ faster than re-reading the document for every question, and the one-time prefill amortizes after about

	extra params	per-doc state	cold st. /doc (s)	online /q (s)	amort. /q (s)	VRAM (GiB)
PAQ-KV (ours)	0	8.15 GiB	11.3	1.04	3.07	44.5
ColQwen2-fresh	2.246 B	11 MiB	2.5	0.46	0.91	21.8
full-context	0	n/a	n/a	11.59	11.59	44.6

Table 2: Deployment profile on an 8-document / 31-query systems subset (frozen configuration, single GPU). "Extra params" = additional retriever/selector parameters beyond the shared reader. "Cold start" is the one-time per-document cost: full-document prefill for PAQ-KV, page-embedding index build for ColQwen2. "Amort. lat./q" adds the cold start amortized at the development pin's observed 5.55 queries/document to the online latency (at the systems subset's own 3.9 queries/document the figures are 3.95/1.10/11.59 s). PAQ-KV's online latency decomposes into 0.19 s selection and 0.85 s masked decode; it is $\approx 2.25\times$ slower and far heavier in resident state than ColQwen2, and $\approx 11\times$ faster than full context (cold start amortized after $N \approx 1.07$ queries/document). PAQ-KV's online latency excludes ≈ 1.99 s/query of removable research-harness input reconstruction (component sum ≈ 3.03 s/query in the current implementation); mean generated lengths differ across arms (5.3/3.5/4.1 tokens for PAQ-KV/ColQwen2/full-context).

one query per document. PAQ-KV therefore does not reduce resident KV memory; it trades retained cache state for reversible query-specific selection. The efficiency claim is simplicity and deployment shape, not speed: no separately trained retriever checkpoint, no stored vector index, one deployed model, and no repeated full-document prefills.

6 Conclusion

PAQ-KV turns a frozen VLM's own KV cache into a queryable, reversible evidence store: prefill once, then per query re-select below-page blocks and decode under a 4D restricted-attention mask over the intact cache, with no gathering and M-RoPE preserved. Per-query re-selection beats committing the identical scorer once in every tested cell, and the improvement over a trained visual retriever at matched budget is confirmed on a sealed set evaluated exactly once ($+0.076$, positive document-clustered lower bound). Four datasets and four readers measure the scope: the advantage reproduces where evidence is dispersed, ties where the cell is conversion-bound, reverses where questions are retrieval-native, and requires a host reader whose own attention is an effective below-page retriever. We offer that scope law as the paper's hypothesis; a sealed-grade confirmation beyond MMLB remains open.

References

Chen, L.; Zhao, H.; Liu, T.; Bai, S.; Lin, J.; Zhou, C.; and Chang, B. 2024. An Image is Worth 1/2 Tokens After Layer 2: Plug-and-Play Inference Acceleration for Large Vision-Language Models. In *European Conference on Computer Vision (ECCV)*. ArXiv:2403.06764; FastV.

Deng, C.; Yuan, J.; Bu, P.; Wang, P.; Li, Z.-Z.; Xu, J.; Li, X.-H.; Gao, Y.; Song, J.; Zheng, B.; and Liu, C.-L. 2025. LongDocURL: a Comprehensive Multimodal Long Document Benchmark Integrating Understanding, Reasoning, and Locating. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*. ArXiv:2412.18424.

Dong, K.; Chang, Y.; Goh, X. D.; Li, D.; Tang, R.; and Liu, Y. 2025a. MMDocIR: Benchmarking Multi-Modal Retrieval for Long Documents. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. ArXiv:2501.08828.

Dong, K.; Chang, Y.; Huang, S.; Wang, Y.; Tang, R.; and Liu, Y. 2025b. Benchmarking Retrieval-Augmented Multimodal Generation for Document Question Answering. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*. ArXiv:2505.16470.

Faysse, M.; Sibille, H.; Wu, T.; Omrani, B.; Viaud, G.; Hudelot, C.; and Colombo, P. 2025. ColPali: Efficient Document Retrieval with Vision Language Models. In *International Conference on Learning Representations (ICLR)*. ArXiv:2407.01449; ColQwen2 is the Qwen2-VL-based variant.

Kim, J.-H.; Kim, J.; Kwon, S.; Lee, J. W.; Yun, S.; and Song, H. O. 2025. KVzip: Query-Agnostic KV Cache Compression with Context Reconstruction. *arXiv preprint arXiv:2505.23416*.

Li, Y.; Huang, Y.; Yang, B.; Venkitesh, B.; Locatelli, A.; Ye, H.; Cai, T.; Lewis, P.; and Chen, D. 2024. SnapKV: LLM Knows What You are Looking for Before Generation. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2404.14469.

Ma, Y.; Zang, Y.; Chen, L.; Chen, M.; Jiao, Y.; Li, X.; Lu, X.; Liu, Z.; Ma, Y.; Dong, X.; Zhang, P.; Pan, L.; Jiang, Y.-G.; Wang, J.; Cao, Y.; and Sun, A. 2024. MMLongBench-Doc: Benchmarking Long-context Document Understanding with Visualizations. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*. ArXiv:2407.01523.

Qwen Team. 2025a. Qwen2.5-VL Technical Report. *arXiv preprint arXiv:2502.13923*.

Qwen Team. 2025b. Qwen3-VL Technical Report. *arXiv preprint arXiv:2511.21631*.

Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; Krueger, G.; and Sutskever, I. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *International Conference on Machine Learning (ICML)*.

Tang, J.; Zhao, Y.; Zhu, K.; Xiao, G.; Kasikci, B.; and Han, S. 2024. Quest: Query-Aware Sparsity for Efficient Long-Context LLM Inference. In *International Conference on Machine Learning (ICML)*. ArXiv:2406.10774.

Wan, Z.; Wu, Z.; Liu, C.; Huang, J.; Zhu, Z.; Jin, P.; Wang, L.; and Yuan, L. 2024. LOOK-M: Look-Once Optimization in KV Cache for Efficient Multimodal Long-Context Inference. In *Findings of the Association for Computational Linguistics: EMNLP*. ArXiv:2406.18139.

Wang, W.; Gao, Z.; Gu, L.; et al. 2025. InternVL3.5: Advancing Open-Source Multimodal Models in Versatility, Reasoning, and Efficiency. *arXiv preprint arXiv:2508.18265*.

Zhang, Z.; Sheng, Y.; Zhou, T.; Chen, T.; Zheng, L.; Cai, R.; Song, Z.; Tian, Y.; Ré, C.; Barrett, C.; Wang, Z.; and Chen, B. 2023. H2O: Heavy-Hitter Oracle for Efficient Generative Inference of Large Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2306.14048.

Zhu, J.; Bao, Y.; Chen, Y.; and Zhu, W. 2026. Attention as Selector: Unlocking VLM Attention for Long Document Page Retrieval. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026, Long Papers)*, 24344–24365. The CAPS method.